Borg: A Complete Open-Source TBDR GPU Pipeline with ASIC Tapeout on 5280 Logic Cells

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Abstract

We present Borg, a Tile-Based Deferred Rendering (TBDR) GPU implemented in Chisel (a Scala-based hardware description language), synthesized with an open-source toolchain, and submitted to physical ASIC fabrication via the Tiny Tapeout programme. The complete rendering pipeline — vertex shader, triangle clipping, edge function rasterizer, texture fetch, Z-buffer, and fragment interpolation — fits within 5,280 logic cells of a Lattice iCE40 UP5K FPGA. To our knowledge, Borg is the first open-source GPU for which a physical ASIC tapeout has been completed. We describe the architecture, design decisions, resource budget, and a path toward hardware-accelerated path tracing.

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1 Introduction

Modern GPU architectures are opaque by design. Every shipped GPU conceals its microarchitecture, instruction set, and physical layout behind NDAs or proprietary binaries. This constrains rendering research in three ways: results are non-reproducible, architectural experiments require expensive EDA licenses, and teaching stops at the API boundary.

Borg is a fully open alternative. Every component — Chisel HDL source, synthesis scripts, place-and-route configuration, and tapeout submission packages — is publicly available at https://github.com/gonsolo/Borg under a permissive licence.

2 Architecture

Borg implements the *Tile-Based Deferred Rendering* (TBDR) paradigm, processing geometry tile-by-tile so that shading is applied only to visible fragments. This approach, also used by PowerVR/Imagination Technologies, Apple GPU, ARM Mali, and Vivante, reduces external memory bandwidth — important for FPGA designs with limited PSRAM throughput.

The pipeline comprises six hardware stages:

(1) **Vertex Shader:** Perspective projection via a hardware reciprocal unit (RCP), executing a custom RISC-V-derived 32-bit ISA.
(2) **Triangle Clipping:** Sutherland-Hodgman clip against the tile viewport.
(3) **Edge Function Unit:** Pineda-style signed area test, one pixel per clock.
(4) **Texture Fetch:** Nearest-neighbour UV sampling from PSRAM via a shared memory controller.
(5) **Fragment Interpolation + Z-Buffer:** Perspective-correct barycentric attribute interpolation with depth test.
(6) **Tile Buffer:** On-chip BRAM with Morton (Z-curve) encoding for cache-friendly 2-D access.

A command FIFO decouples the CPU from the rendering pipeline. A SystemRDL register map (generated via PeakRDL-chisel) provides a clean software interface.

3 Resource Budget

Table 1: iCE40 UP5K resource utilisation (pico-ice board).

Resource	Used	Available
Logic Cells	∼5 100	5 280
4 kb BRAMs	[N]	30
DSPs	0	8

The iCE40 UP5K provides no DSP blocks usable for FP multiply-add; all floating-point computation is implemented in soft logic.

4 ASIC Tapeouts

The GF180 tapeout was funded by the Google Open-Source Silicon Programme; TTIHP26a and TTSKY26a via the Tiny Tapeout programme. Chips for TTIHP26a are expected in July 2026.

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Table 2: Physical tapeouts submitted as part of the Borg project.

Run	Process	Content	Submitted
GF180	GF 180 nm	Diffuse shader ASIC	2022
TTIHP26a	IHP 130 nm	FP adder peripheral	Mar 2026
TTSKY26a	SKY130	Triangle rasteriser	May 2026

5 Comparison with Related Open GPU Projects

Nyuzi [1] is an open GPGPU targeting compute workloads (no rasterisation pipeline, no tapeout). Vortex [3] is a RISC-V GPGPU targeting research on GPU computing; it targets Xilinx FPGAs and has not been taped out. Borg differs in focus (graphics, not compute), architecture (TBDR, not SIMD lanes), and, critically, in having submitted physical silicon.

6 Current Status and Future Work

Phase 1 (Steps 1–20: the full rendering pipeline) is complete. Phase 2 (GPU autonomy — DMA engine, write port, vertex sequencer) is in progress and will remove CPU orchestration from the render loop. Phase 3 plans a Linux-capable CPU integration, and Phase 4 targets a minimal Mesa Vulkan driver.

The longer-term goal is integration with Gonzales [4], a production path tracer, to create the first fully open-source hardware path tracing system.

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[1] Jeff Bush. 2015. *Nyuzi: Open Source GPGPU Processor*. https://github.com/jbush001/NyuziProcessor

[2] Matt Pharr, Wenzel Jakob, and Greg Humphreys. 2023. Physically Based Rendering: From Theory to Implementation. MIT Press.

[3] Blaise Tine, Krishna Patel, Aniruddh Ramesh, Kyle Daruwalla, and Hyesoon Kim. 2021. Vortex: Extending the RISC-V ISA for GPGPU and 3D-Graphics. In *International Symposium on Microarchitecture (MICRO)*.

[4] Andreas Wendleder. 2024. *Gonzales Renderer*. https://gonsolo.github.io/gonzales/00_index.html Online book documenting the Gonzales path tracer, structurally similar to Physically Based Rendering [2].